

Inspect a user-process page table

Each Page table entries have a virtual address, page table entry as hex code, physical address, permission bits.

First 10-page table entries

1. This is a user page, it is readable and executable but not writable, it has an accessed bit set, likely a code page.
2. This is a user page, it is readable and executable but not writable, it has an accessed bit set, likely a code page.
3. This is a user page, it is readable and writable but not executable.
4. This is a kernel page, it is readable and writable, but not executable, could be some kernel data
5. This is a user page, it is readable and writable but not executable, but the accessed and dirty bit has been set so this page has been read and written. Likely user heap

From pages 6 to 10 all are unmapped

Last 10-page table entries

From pages 1 to 8 all are unmapped

9. This is a kernel page, it is readable and writable, its access and dirty bits are set. Most likely a kernel stack
10. This is a kernel page, it is readable and executable but not writable, can be kernel code.

Below is the attached output I have received.

AmmarKhan-Mint-OS [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld
riscv64-unknown-elf-objdump -S user/_forphan > user/forphan.
riscv64-unknown-elf-objdump -t user/_forphan | sed '1,/SYMBOL
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attribute
in-strlen -fno-builtin-memset -fno-builtin-memmove -fno-buil
iltin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld
riscv64-unknown-elf-objdump -S user/_dorphan > user/dorphan.
riscv64-unknown-elf-objdump -t user/_dorphan | sed '1,/SYMBOL
riscv64-unknown-elf-gcc -Wall -Werror -Wno-unknown-attribute
in-strlen -fno-builtin-memset -fno-builtin-memmove -fno-buil
iltin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld
riscv64-unknown-elf-objdump -S user/_pgtbltest > user/pgtblt
riscv64-unknown-elf-objdump -t user/_pgtbltest | sed '1,/SYM
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest
ser/_dorphan user/_pgtbltest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitma
balloc: first 984 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/

xv6 kernel is booting

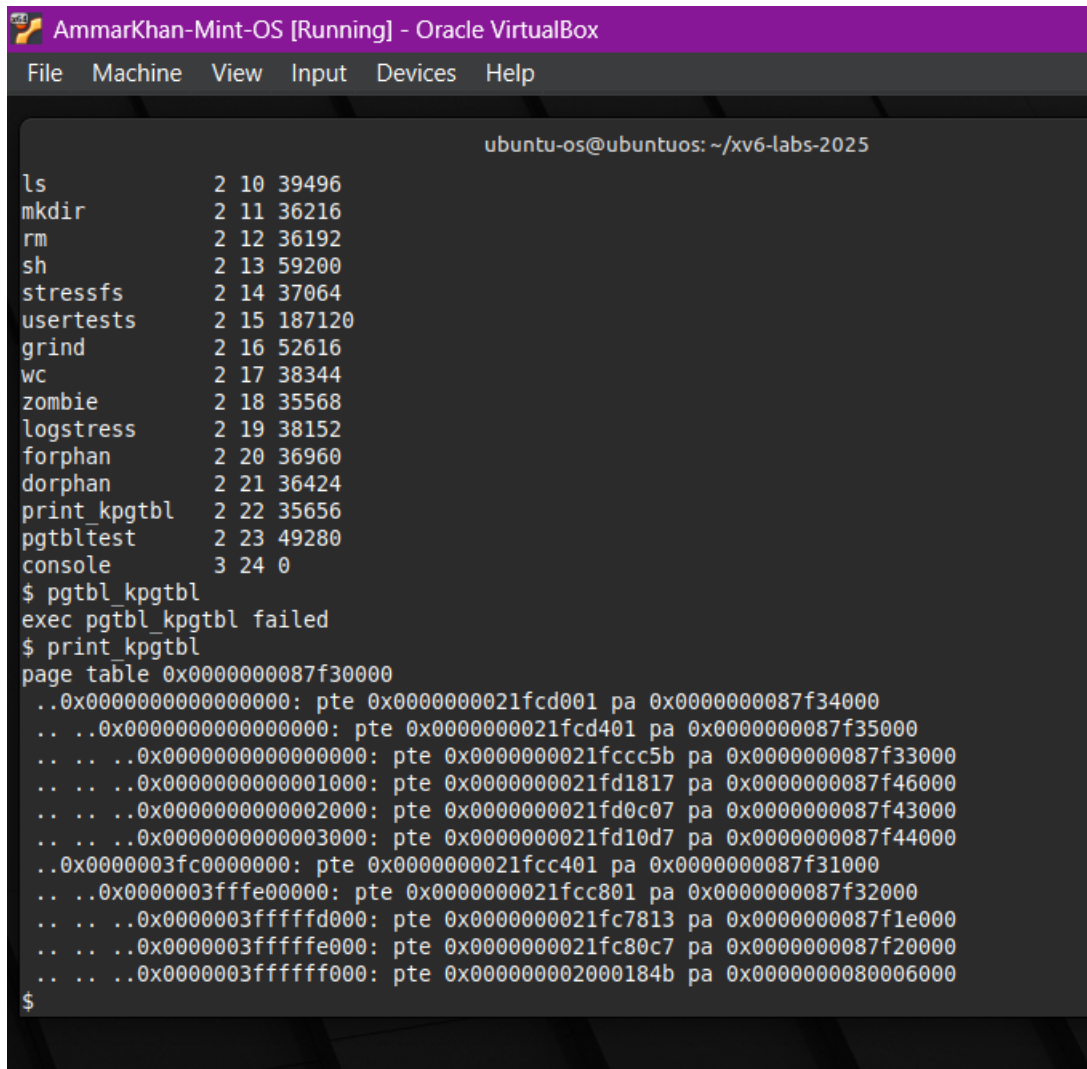
hart 1 starting
hart 2 starting
init: starting sh
$ pgtbltest
print_pgtbl starting
va 0x0 pte 0x21FC885B pa 0x87F22000 perm 0x5B
va 0x1000 pte 0x21FC7C5B pa 0x87F1F000 perm 0x5B
va 0x2000 pte 0x21FC7817 pa 0x87F1E000 perm 0x17
va 0x3000 pte 0x21FC7407 pa 0x87F1D000 perm 0x7
va 0x4000 pte 0x21FC70D7 pa 0x87F1C000 perm 0xD7
va 0x5000 pte 0x0 pa 0x0 perm 0x0
va 0x6000 pte 0x0 pa 0x0 perm 0x0
va 0x7000 pte 0x0 pa 0x0 perm 0x0
va 0x8000 pte 0x0 pa 0x0 perm 0x0
va 0x9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF6000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFD000 pte 0x0 pa 0x0 perm 0x0
va 0x3FFFFFFE000 pte 0x21FD08C7 pa 0x87F42000 perm 0xC7
va 0x3FFFFFFF000 pte 0x2000184B pa 0x80006000 perm 0x4B
print_pgtbl: OK
ugetpid_test starting
usertrap(): unexpected scause 0xd pid=3
sepc=0x79e stval=0x3ffffd000
```

Q2

uptime(), As the kernel can periodically update the current tick count using the shared USYSCALL and the user space can read it directly without syscall

time(), As the kernel periodically writes time to the shared memory

Q3



```
ubuntu-os@ubuntuos: ~/xv6-labs-2025

ls          2 10 39496
mkdir       2 11 36216
rm          2 12 36192
sh          2 13 59200
stressfs    2 14 37064
usertests   2 15 187120
grind       2 16 52616
wc          2 17 38344
zombie      2 18 35568
logstress   2 19 38152
forphan     2 20 36960
dorphan     2 21 36424
print_kpgtbl 2 22 35656
pgtbltest   2 23 49280
console     3 24 0
$ pgtbl_kpgtbl
exec pgtbl_kpgtbl failed
$ print_kpgtbl
page table 0x0000000087f30000
..0x0000000000000000: pte 0x0000000021fcd001 pa 0x0000000087f34000
.. ..0x0000000000000000: pte 0x0000000021fcd401 pa 0x0000000087f35000
.. .. ..0x0000000000000000: pte 0x0000000021fccc5b pa 0x0000000087f33000
.. .. ..0x00000000000001000: pte 0x0000000021fd1817 pa 0x0000000087f46000
.. .. ..0x00000000000002000: pte 0x0000000021fd0c07 pa 0x0000000087f43000
.. .. ..0x00000000000003000: pte 0x0000000021fd10d7 pa 0x0000000087f44000
..0x00000003fc0000000: pte 0x0000000021fcc401 pa 0x0000000087f31000
.. ..0x00000003fffe00000: pte 0x0000000021fcc801 pa 0x0000000087f32000
.. .. ..0x00000003fffffd000: pte 0x0000000021fc7813 pa 0x0000000087f1e000
.. .. ..0x00000003fffffe000: pte 0x0000000021fc80c7 pa 0x0000000087f20000
.. .. ..0x00000003ffffff000: pte 0x000000002000184b pa 0x0000000080006000
$
```